This document shows sample runs of the forward step and the backward step of Gaussian Elimination, with automatically-generated explanations of what is happening at every step. Your output does not need to match exactly, but it should show the same matrix transformations.

EXAMPLE 1. FORWARD STEP

Starting matrix (as a list of lists): `[[0, 0, 1, 0, 2], [1, 0, 2, 3, 4], [3, 0, 4, 2, 1], [1, 0, 1, 1, 2]]`

```
The matrix is currently:
[[0. 0. 1. 0. 2.]
 [1. 0. 2. 3. 4.]
 [3. 0. 4. 2. 1.]
 [1. 0. 1. 1. 2.]]
Now looking at row 0
Swapping rows 0 and 1 so that entry 0 in the current row is non-zero
The matrix is currently:
[[1. 0. 2. 3. 4.]
 [0. 0. 1. 0. 2.]
 [3. 0. 4. 2. 1.]
 [1. 0. 1. 1. 2.]]
Adding row 0 to rows below it to eliminate coefficients in column 0
The matrix is currently:
[[  1.   0.   2.   3.   4.]
 [  0.   0.   1.   0.   2.]
 [  0.   0.  -2.  -7. -11.]
 [  0.   0.  -1.  -2.  -2.]]
================================================================================
Now looking at row 1
Swapping rows 1 and 1 so that entry 2 in the current row is non-zero
The matrix is currently:
[[  1.   0.   2.   3.   4.]
 [  0.   0.   1.   0.   2.]
 [  0.   0.  -2.  -7. -11.]
 [  0.   0.  -1.  -2.  -2.]]
Adding row 1 to rows below it to eliminate coefficients in column 2
The matrix is currently:
[[ 1.  0.  2.  3.  4.]
 [ 0.  0.  1.  0.  2.]
 [ 0.  0.  0. -7. -7.]
 [ 0.  0.  0. -2.  0.]]
================================================================================
Now looking at row 2
Swapping rows 2 and 2 so that entry 3 in the current row is non-zero
The matrix is currently:
[[ 1.  0.  2.  3.  4.]
 [ 0.  0.  1.  0.  2.]
 [ 0.  0.  0. -7. -7.]
 [ 0.  0.  0. -2.  0.]]
Adding row 2 to rows below it to eliminate coefficients in column 3
The matrix is currently:
[[ 1.  0.  2.  3.  4.]
 [ 0.  0.  1.  0.  2.]
 [ 0.  0.  0. -7. -7.]
 [ 0.  0.  0.  0.  2.]]
```

This lab has been adapted from one used in previous years by Prof. Michael Guerzhoy.

```
================================================================================
Now looking at row 3
Swapping rows 3 and 3 so that entry 4 in the current row is non-zero
The matrix is currently:
[[ 1.  0.  2.  3.  4.]
 [ 0.  0.  1.  0.  2.]
 [ 0.  0.  0. -7. -7.]
 [ 0.  0.  0.  0.  2.]]
Adding row 3 to rows below it to eliminate coefficients in column 4
The matrix is currently:
[[ 1.  0.  2.  3.  4.]
 [ 0.  0.  1.  0.  2.]
 [ 0.  0.  0. -7. -7.]
 [ 0.  0.  0.  0.  2.]]
================================================================================
Done with the forward step
The matrix is currently:
[[ 1.  0.  2.  3.  4.]
 [ 0.  0.  1.  0.  2.]
 [ 0.  0.  0. -7. -7.]
 [ 0.  0.  0.  0.  2.]]
```

EXAMPLE 2. FORWARD STEP AND BACKWARD STEP

Starting matrix (the augmented matrix for the system in Question 8): `[[1, -2, 3, 22], [3, 10, 1, 314], [1, 5, 3, 92]]`

```
The matrix is currently:
[[  1  -2   3  22]
 [  3  10   1 314]
 [  1   5   3  92]]
================================================================================
Now performing the forward step
Now looking at row 0
Swapping rows 0 and 0 so that entry 0 in the current row is non-zero
The matrix is currently:
[[  1  -2   3  22]
 [  3  10   1 314]
 [  1   5   3  92]]
Adding row 0 to rows below it to eliminate coefficients in column 0
The matrix is currently:
[[  1.  -2.   3.  22.]
 [  0.  16.  -8. 248.]
 [  0.   7.   0.  70.]]
================================================================================
Now looking at row 1
Swapping rows 1 and 1 so that entry 1 in the current row is non-zero
The matrix is currently:
[[  1.  -2.   3.  22.]
 [  0.  16.  -8. 248.]
 [  0.   7.   0.  70.]]
Adding row 1 to rows below it to eliminate coefficients in column 1
The matrix is currently:
[[  1.   -2.    3.   22. ]
 [  0.   16.   -8.  248. ]
 [  0.    0.    3.5 -38.5]]
```

This lab has been adapted from one used in previous years by Prof. Michael Guerzhoy.

```
================================================================================
Now looking at row 2
Swapping rows 2 and 2 so that entry 2 in the current row is non-zero
The matrix is currently:
[[  1.   -2.    3.   22. ]
 [  0.   16.   -8.  248. ]
 [  0.    0.    3.5 -38.5]]
Adding row 2 to rows below it to eliminate coefficients in column 2
The matrix is currently:
[[  1.   -2.    3.   22. ]
 [  0.   16.   -8.  248. ]
 [  0.    0.    3.5 -38.5]]
================================================================================
Done with the forward step
The matrix is currently:
[[  1.   -2.    3.   22. ]
 [  0.   16.   -8.  248. ]
 [  0.    0.    3.5 -38.5]]
================================================================================
Now performing the backward step
Adding row 2 to rows above it to eliminate coefficients in column 2
The matrix is currently:
[[  1.   -2.    0.   55. ]
 [  0.   16.    0.  160. ]
 [  0.    0.    3.5 -38.5]]
================================================================================
Adding row 1 to rows above it to eliminate coefficients in column 1
The matrix is currently:
[[  1.    0.    0.   75. ]
 [  0.   16.    0.  160. ]
 [  0.    0.    3.5 -38.5]]
================================================================================
Adding row 0 to rows above it to eliminate coefficients in column 0
The matrix is currently:
[[  1.    0.    0.   75. ]
 [  0.   16.    0.  160. ]
 [  0.    0.    3.5 -38.5]]
================================================================================
Now dividing each row by the leading coefficient
The matrix is currently:
[[  1.   0.   0.  75.]
 [  0.   1.   0.  10.]
 [  0.   0.   1. -11.]]
================================================================================
```